

EXPERIMENT 8 TCP/UDP sockets

Implement echo client using TCP/UDP sockets using Java Program

AIM:

To implement an Echo Client-Server model using TCP/UDP sockets in Java.,To establish communication between client and server and verify message echo functionality.

PROCEDURE :

1. Write the server program using TCP/UDP sockets to listen on a specific port and wait for client requests.
2. Implement the client program to connect to the server (TCP) or send datagrams (UDP) with user input messages.
3. In the server, receive the message from the client and send the same message back (echo).
4. Compile and run the server program first, then execute the client program in a separate terminal.
5. Test the application by sending messages from the client and verify the echoed response from the server.
- 6.

OUTPUT :

```
C:\Users\Tcs\Desktop>java main
Hello from Notepad!
```

```
T3server.java
File Edit View

import java.net.*;
import java.io.*;
import java.util.*;
class T3Server {
    public static void main(String[] args) {
        try
        {
            ServerSocket ss =new ServerSocket(5000);
            System.out.println("waiting for Client");
            Socket server = ss.accept();
            System.out.println("Server is connected");
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

```
Command Prompt
Microsoft Windows [Version 10.0.22631.5039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Tcs>cd desktop

C:\Users\Tcs\Desktop>javac T3Client.java

C:\Users\Tcs\Desktop>java T3Client
Client is connected

C:\Users\Tcs\Desktop>

C:\Users\Tcs\Desktop>javac T3Server.java

C:\Users\Tcs\Desktop>java T3Server
waiting for Client
█
```

RESULT :

The Echo Client-Server application was successfully implemented using TCP/UDP sockets in Java.